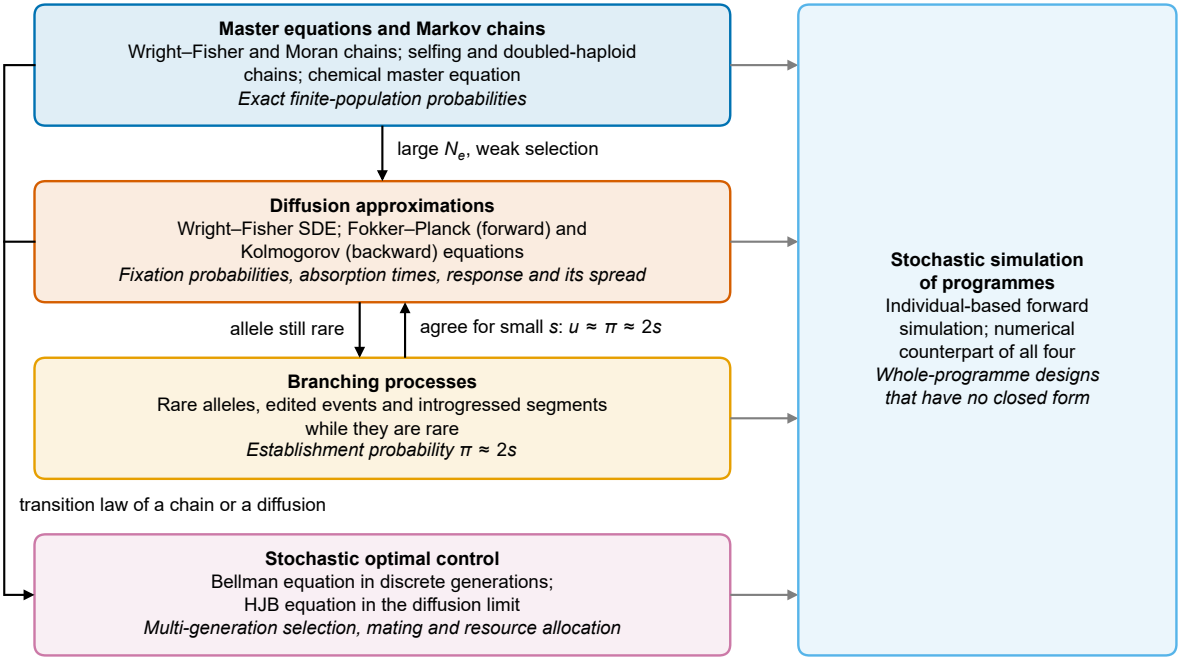

Master equations and Markov chains
Wright–Fisher and Moran chains; selfing and doubled-haploid chains; chemical master equation
Exact finite-population probabilities
large $N_e$, weak selection
Diffusion approximations
Wright–Fisher SDE; Fokker–Planck (forward) and Kolmogorov (backward) equations
Fixation probabilities, absorption times, response and its spread
allele still rare
agree for small $s$: $u \approx \pi \approx 2s$
Branching processes
Rare alleles, edited events and introgressed segments while they are rare
Establishment probability $\pi \approx 2s$
transition law of a chain or a diffusion
Stochastic optimal control
Bellman equation in discrete generations; HJB equation in the diffusion limit
Multi-generation selection, mating and resource allocation
Stochastic simulation of programmes
Individual-based forward simulation; numerical counterpart of all four
Whole-programme designs that have no closed form